Review: Equations of lines

1 Explain why the following statements are incorrect.

a "A line with an undefined slope has no equation."

b "The slope of the line with equation $7x + y = 3$ is 7."

2 A line has the equation $y = 6\,(3x - 2)$.

a Rewrite $y = 6\,(3x - 2)$ in the form $y = mx + c$.

b State the slope and y-intercept of the equation.

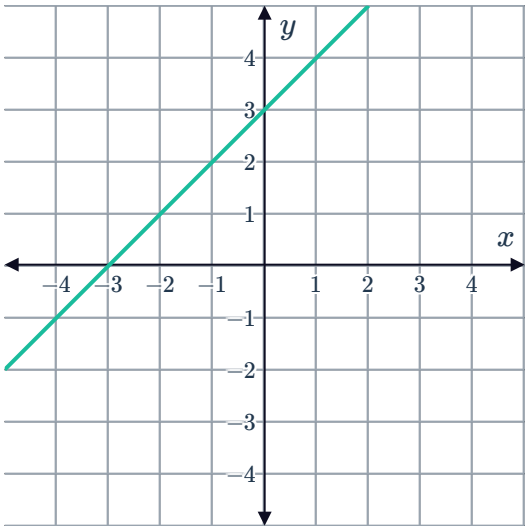
3 Write the equation of the following lines in slope-intercept form:

a A line whose slope is -8 and crosses the y-axis at -9.

b A line that has a slope equal to -7, and cuts the y-axis at 7.

c A line that passes through the point $A\,(-2, -9)$ and has a slope of -2.

4 The following is a graph of $y = mx + b$. Sketch the graph of y when b is decreased by 5.



5 Line L_1 has the following equation: $y = x - 2$

a Find the y-value of the y-intercept of the line.

b Find the x-value of the x-intercept of the line.

c Find the y-coordinate of a point that has an x-coordinate of -5.

d Sketch the line $y = x - 2$ on the coordinate plane.

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6 Express the following equations in standard form.

a $y - 5x = 5$

b $y = 2x - 5$

c $y = \dfrac{2x}{5} + 4$

7 A line has slope $\dfrac{14}{3}$ and passes through the point $(-5, -10)$.

a Find the value of b for this line.

b Write the equation of the line in standard form.

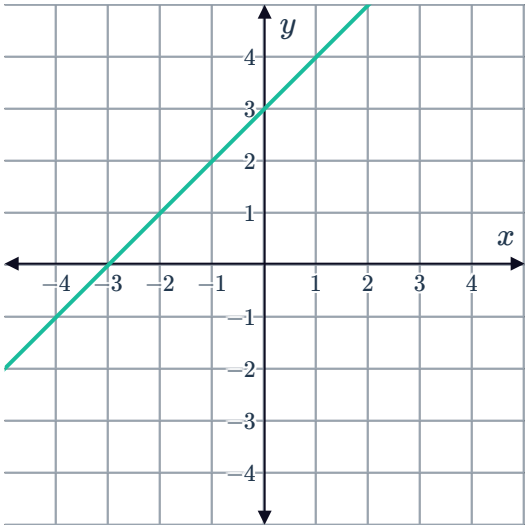
8 Determine whether the following could be the equation of the graphed line.

a $2x - 2y = 6$

b $2x - 2y = -6$

c $2x + 2y = 6$

d $2x + 2y = -6$



9 $A\,(-1, 4)$ is a point on a straight line. If the slope of the line is 4, determine whether the following statements are correct.

a $\dfrac{x+1}{y-4} = 4$

b $\dfrac{y-4}{x+1} = 4$

c $\dfrac{y+1}{x-4} = 4$

d $\dfrac{4-y}{-1-x} = 4$

10 A line passes through the points $(3, -5)$ and $(-7, 2)$.

a Find the slope of the line.

b Find the equation of the line in slope-intercept or standard form.

11 Use the two point formula to derive the equation of the line that passes through the points $A\,(8, 1)$ and $B\left(-2, \dfrac{1}{7}\right)$. Give your answer in standard form.

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12 A line has an x-intercept of 8 and a y-intercept of 5.
By considering the intercept form of a straight line, write the equation of the line in the standard form.

13

a Find the equation, in standard form, of the line that passes through $A\,(7, -14)$ and $B\,(-7, -15)$.

b Find the x-coordinate of the point of intersection of the line that goes through A and B, and the line $y = x - 8$.

c Hence, find the y-coordinate of the point of intersection.

Additional questions

14 For each of the following linear equations:

i State the value of m, the slope.

ii State the value of b, the y-intercept.

a $y = 6x + 4$

b $y = -3x - 8$

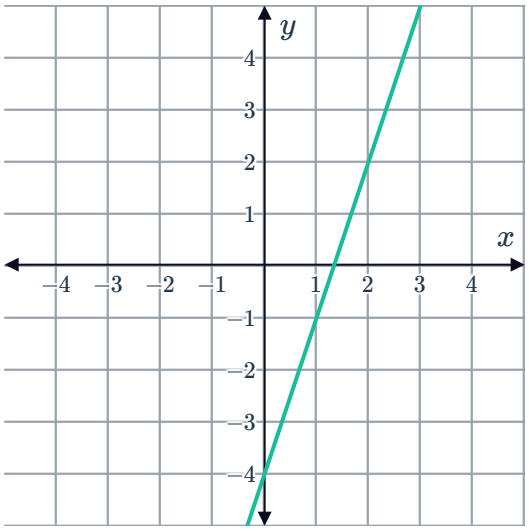
15 For each of the following graphs:

i Determine the slope of the line.

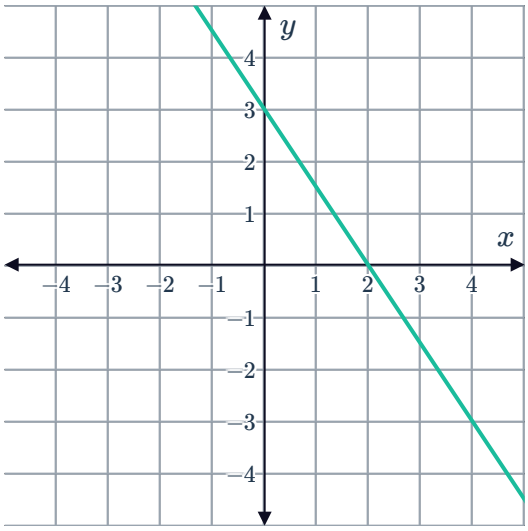
ii Determine the y-intercept of the line.

iii Determine the equation of the line.

a



b



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16 Consider the following linear equations:

- Line 1: $y = 2 + 6x$
- Line 2: $y = 5x + 6$
- Line 3: $y = 6x$
- Line 4: $y = \frac{x}{6} + 2$
- Line 5: $y = 5 - 6x$
- Line 6: $y = 6x - 2$

State the three lines that have the same slope.

17 For each of the following lines, state whether the following lines have a y-intercept or not. If the line does have a y-intercept, state the coordinates of it.

a $y = 2$ **b** $5x = 4y$ **c** $y = -4x + 1$ **d** $x = 1$

18 Suppose the Line L_1 has the equation $y = \frac{-9x + 3}{3}$.

a Simplify L_1 to the form $y = mx + b$.

b State the slope of the line.

c Find the y-value of the y-intercept of the line.

19 Determine whether each of the following lines has the same slope as $y = 6x + 3$ or not.

a $y = 6x$ **b** $y = 3x + 3$ **c** $12x - 2y = 3$ **d** $18x + 3y = 0$

20 Find the equation of the lines matching the following descriptions:

a A line that has the same slope as $y = 2 - 4x$ and the same y-intercept as $y = -7x - 3$.

b A line whose slope is $\frac{2}{3}$ and goes through the point $(0, 5)$.

c A line that is horizontal and goes through the point $(9, -2)$.

d A line that is vertical and goes through the point $(11, -9)$.

21 Sketch the following lines on a coordinate plane:

a The line that has a y-intercept of -4 and whose slope is $-\frac{4}{3}$

b The line $y = 2x + 4$

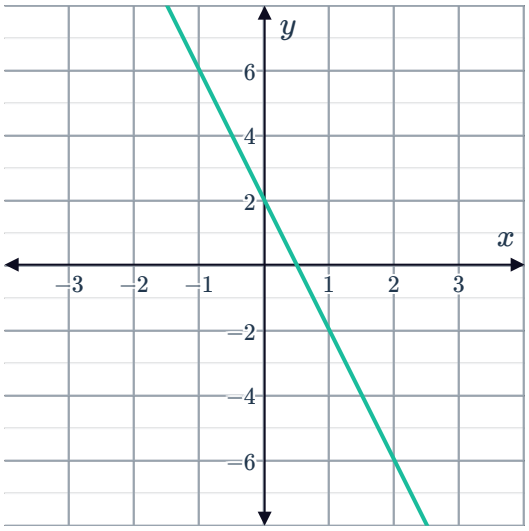
c The line $y = -2x + 4$

d The line $y = \frac{1}{2}x - 1$

e The line $x = -3$

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22 Consider the following line:



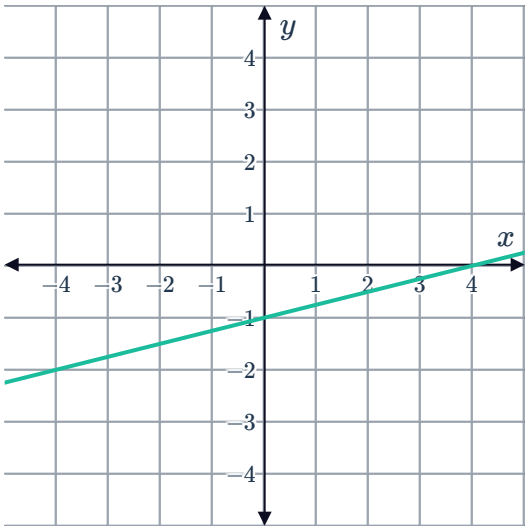
a Copy and complete the following table of values.

x	-1	0	1	2
y				

b State the values of the slope, m and the y-intercept b.

c Write the linear equation expressing the relationship between x and y.

23 Consider the following line:



a State the values of the slope, m and the y-intercept b.

b Write the linear equation expressing the relationship between x and y.

24 For the following lines passing through the given two points:

i Find the slope of the line.

ii Find the equation of the line in the form $y = mx + b$.

a $(0, 5)$ and $(2, 9)$

b $(-3, 9)$ and $(4, 2)$

c $(-6, 8)$ and $(6, 0)$

d $(-7, 3)$ and $(-7, -2)$

25 State the standard form of a linear equation.

26 State which of the following linear equations are in standard form:

a $2x - 5y = 0$

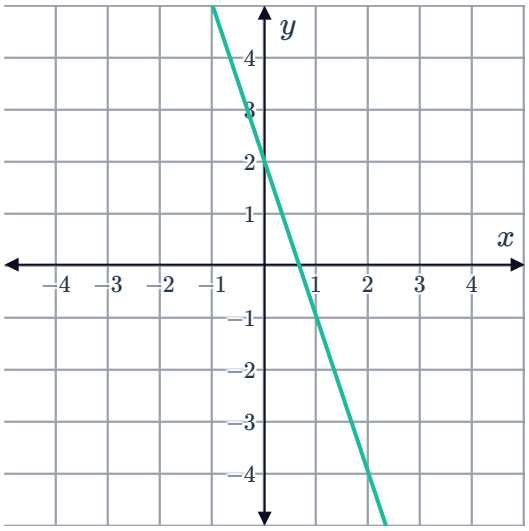
b $x = 10$

c $y = \frac{1}{2}x - 5$

d $-3x + \frac{7}{8}y = 11$

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27 Consider the following line:



- **a** State the y-intercept.
- **b** Find the slope, m, of the line.
- **c** Find the equation of the line in the form $y = mx + b$.
- **d** Rewrite the equation of the line in standard form.

28 State the point-slope form of a linear equation.

29 For each of the following equations, state the form they are written in:

- **a** $y = -4x + 5$
- **b** $y - 1 = 5(x - 4)$
- **c** $5x - 3y = 8$

30 Write the equation of the following lines in point-slope form:

- **a** A line passing through the point $(7, 1)$ has a slope of 4.
- **b** A line passing through the point $(4, 0)$ has a slope of -5.
- **c** A line passing through the point $(1, 6)$ has a slope of 0.

31 For each of the following lines:

- **i** Find the slope, m, of the line.
- **ii** Write the equation of the line in point-slope form.

- **a** A line passes through the two points $(7, 6)$ and $(9, 12)$.
- **b** A line passes through the two points $(4, -8)$ and $(2, 0)$.

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32 A line has slope $-\frac{3}{2}$ and passes through the point $(2, -2)$.

a Find the value of b for this line.

b Hence, write the equation of the line in slope-intercept form.

c Derive the same equation by using the point-slope formula.

d Sketch the graph of the line.

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